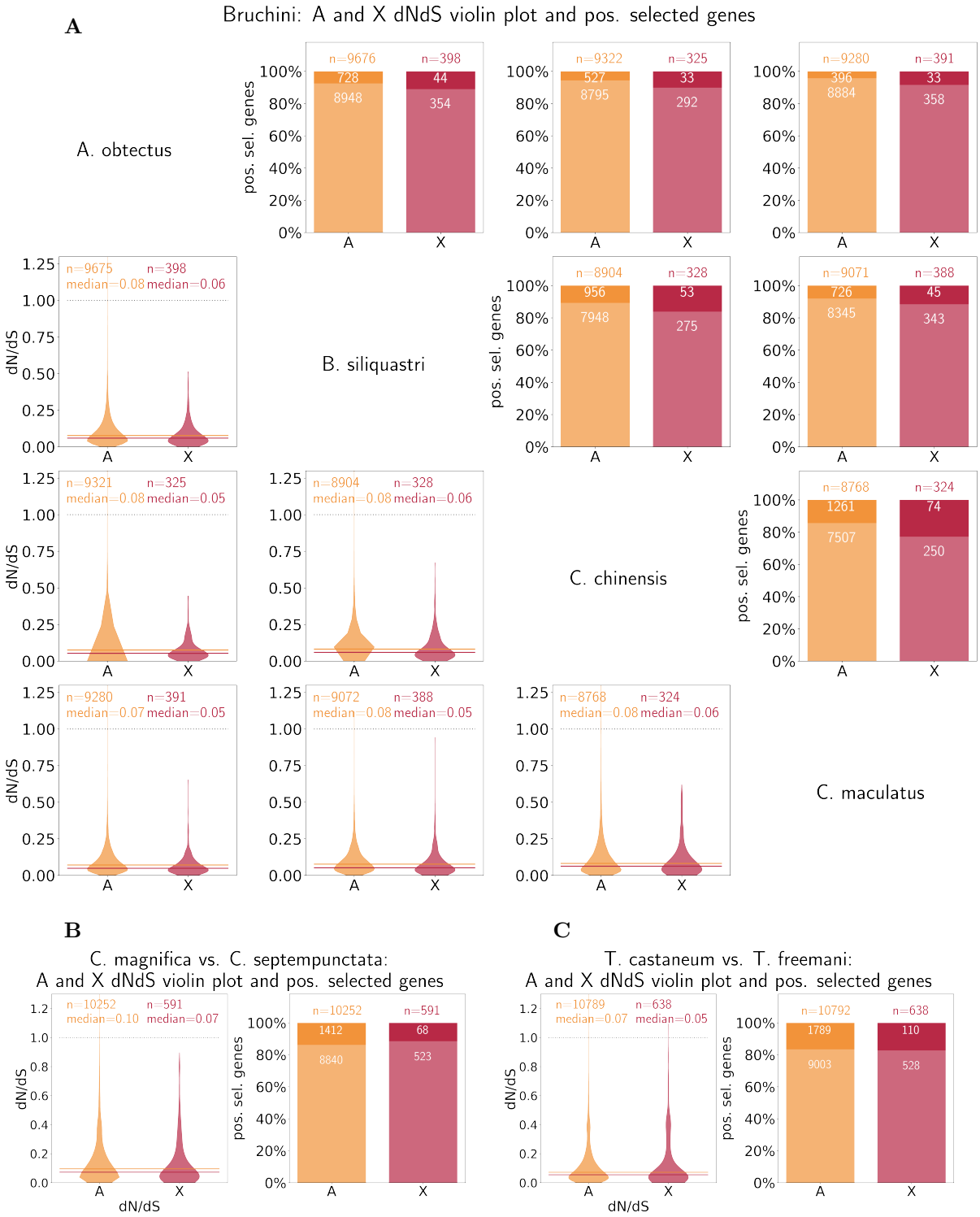

Figure 1: **d_N/d_S and abundance of positively selected genes.** *Violin plots*: Pairwise comparisons between members of three species groups with violin plots of d_N/d_S values for X-linked and autosomal 1-to-1 orthologs, and scatterplots of d_S vs. d_N. Permutation tests (n=10000) show that d_N/d_S is significantly lower for X-linked orthologs in all comparisons within *Bruchini* (**A**) and *Coccinella* (**B**), but not *Tribolium* (**C**) (see Fig. S7 for permutation test results). *Bar plots*: The bar plot show the percentage of genes that have codons under positive selection according to paml site-model analysis (likelihood ratio test comparison between paml site models M1a and M2a). Permutation tests (n=10000) show that *Bruchini* (**A**) have an enrichment of positively selected X-linked orthologs in all pairwise comparisons, while *Coccinella* (**B**) and *Tribolium* (**C**) show no significant difference (see Fig. S8).

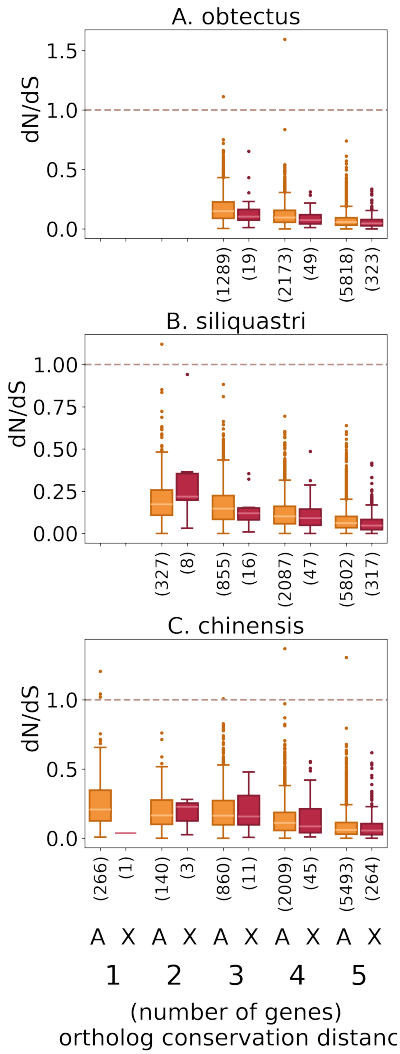

Figure 2: **Gene age**: X-axes ticks show sample size in parentheses, as well as gene age (1 to 5), and autosomal or X-linkage of orthologs (A or X). The box plots show d_N/d_S for autosomal (yellow) or X-linked (red) orthologs by gene age for *C. maculatus* compared to three other textitBruchini species.

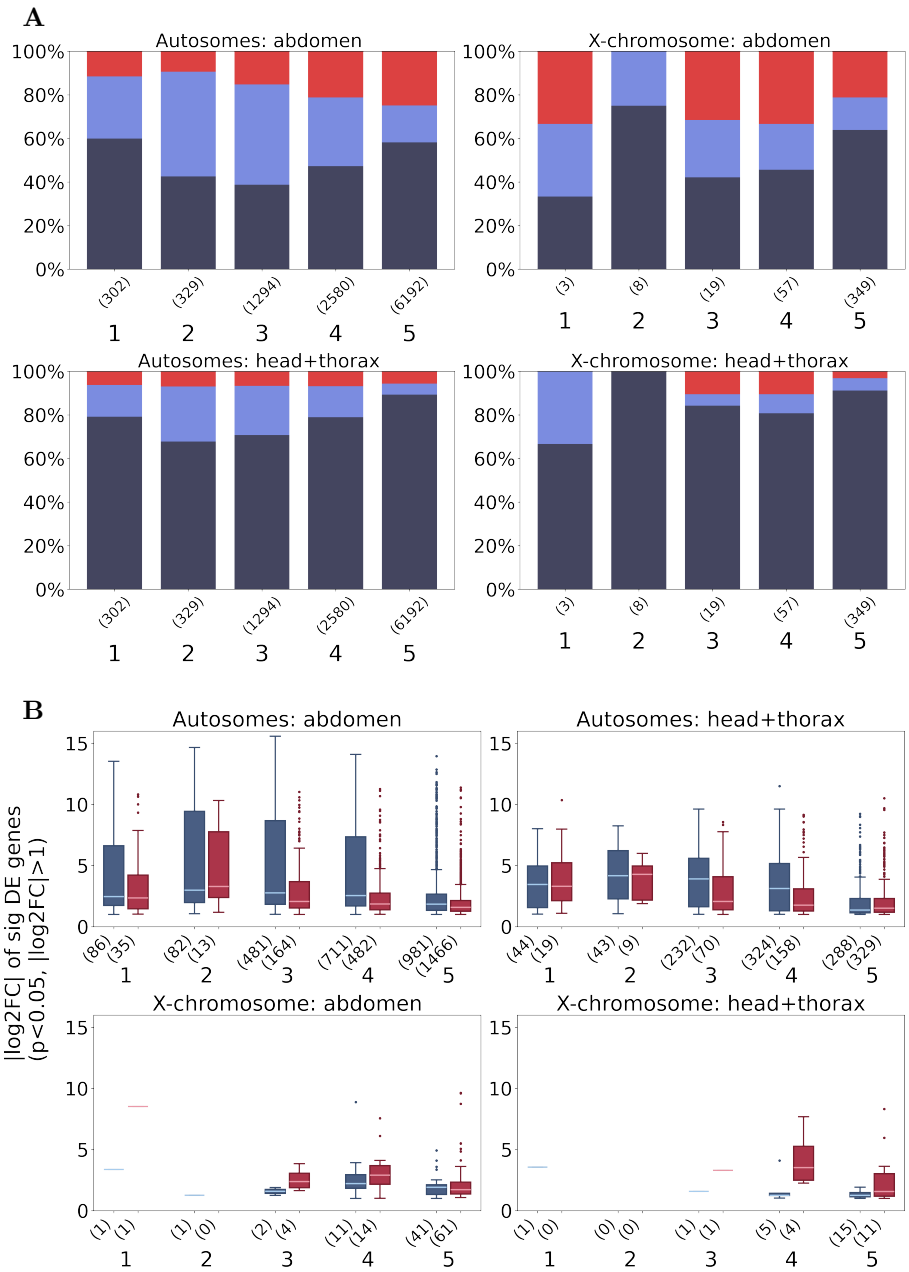

Figure 3: **gene age and sex-bias analysis**: X-axes ticks show sample size in parentheses, as well as gene age (1 to 5), and autosomal or X-linkage of orthologs (A or X). (**A**): proportion of sex-bias categories for each gene age category according to abdominal or head and thorax tissues, as well as autosomal or X-linked orthologs. Unbiased in grey, female-biased in red and male-biased in blue. (**B**): Absolute log2FC for significantly male-biased (blue) or female-biased (red) orthologs of *C. maculatus* and *C. chinensis* for increasing gene age. Split by autosomal and X-linked orthologs, as well as log2FC determined from abdominal tissue (reproductive) or head and thorax tissue (somatic).

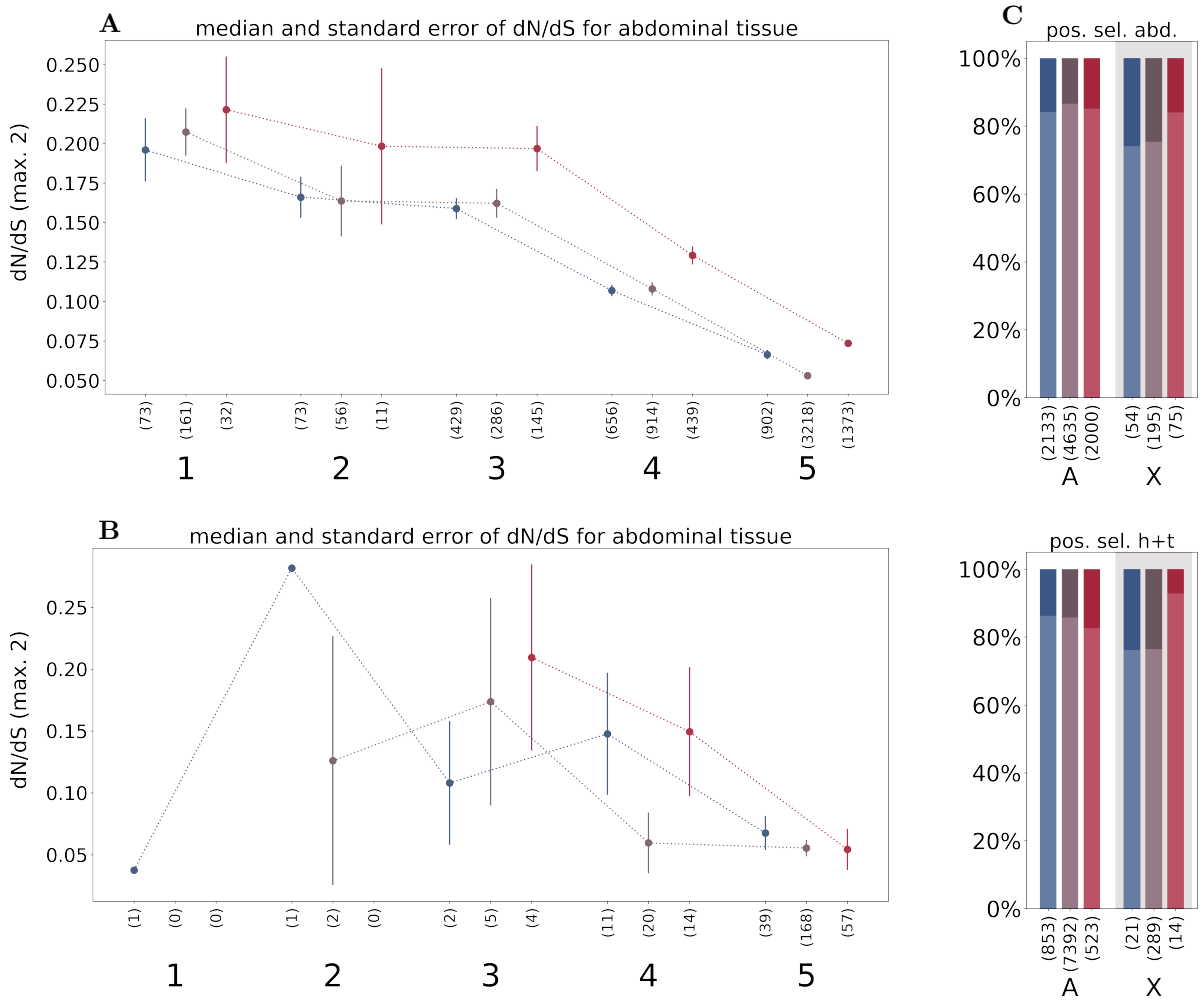

Figure 4: **gene age and sex-bias analysis**: X-axes ticks show sample size in parentheses, as well as gene age (1 to 5), or autosomal or X-linkage of orthologs (A or X). (**A**,**B**): median $d_N d_S$ and standard error of autosomal orthologs, split by gene age as well as sex-bias category according to abdominal tissues with male-bias being blue, unbiased being grey and female-biased being red. **A**: autosomal, **B**: X-linked. (**C**): Proportion of genes under positive selection according to paml site model analysis (darker section) split by Autosomal and X-linked orthologs, as well as male-biased (blue), unbiased (grey) and female-biased (red) orthologs according to either abdominal tissue (reproductive) or head and thorax tissue (somatic).